Alex Crespillo López

Predoctoral researcher in hydrological drought and climatology

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I study how hydrological drought starts in some reaches of a river network and propagates to others. I developed an event-based methodology that reconstructs this propagation between nested gauging stations, published in *Journal of Hydrology* and applied at continental scale in Europe and the United States. My work combines large-scale analysis of streamflow series, distributed hydrological modeling and open scientific programming.

CURRENT POSITION

Predoctoral researcher — IPE-CSIC, Laboratory of Climatology and Climate Services (LCSC, lcsc.csic.es), Zaragoza · since January 2025

- Predoctoral contract of the Momentum programme (CSIC).
- PhD candidate at the Faculty of Physics, University of Barcelona. Supervisor: Sergio M. Vicente-Serrano. Expected defence: December 2028.
- Thesis in four parts: the event-based tracking methodology (published), changes in drought propagation over recent decades (in preparation), the physical mechanisms that control it, and its prediction.

RESEARCH EXPERIENCE

Predoctoral researcher — IPE-CSIC, Zaragoza · January 2025 – present

- Development of an event-based method that reconstructs hydrological drought propagation between nested gauging stations from long series of daily streamflow.
- Collaboration with the University of Minnesota (Alexander Michalek) since June 2026 on the evaluation of the distributed hydrological model Tiger-HLM under drought conditions.
- Contribution to the group's climate services on drought, dry spells and extreme precipitation in Spain.
- Release of the method's code as open source (Zenodo, MIT licence).

Master's internship — Andorra Recerca + Innovació (AR+I), Andorra la Vella · September 2023 – February 2024

- Analysis of meteorological and climate data, support for decision-making on environmental and air-pollution measures, and research on atmospheric dynamics in areas of complex orography.

EDUCATION

MSc in Mathematical Engineering and High-Performance Computing (HPC) — UNIR · 2025 – ongoing

MSc in Meteorology — University of Barcelona · 2022 – 2024

- Master's thesis: "Analysis of thermal inversion episodes in the central valley of Andorra and their relationship with meteorological variables and pollutants". Grade: 8.5/10.

BSc in Geology — Autonomous University of Barcelona · 2018 – 2022 · specialisation in Environmental Geology

- Bachelor's thesis: "Characterisation of native hydrogen reservoirs in the Iberian Peninsula".

PUBLICATIONS

JOURNAL ARTICLES

1. **Crespillo López, A.**, Vicente-Serrano, S. M., Gimeno, L. (2026). *Tracking hydrological drought propagation in river networks: an event-based framework.* Journal of Hydrology, 680, 136544. https://doi.org/10.1016/j.jhydrol.2026.136544
2. Vicente-Serrano, S. M., Beguería, S., Royo, A., El Kenawy, A., Franquesa, M., Halifa-Marín, A., Adell-Michavila, M., **Crespillo, A.**, Pérez-Pajuelo, D., Domínguez-Castro, F., Azorin-Molina, C., Gimeno, L., Nieto, R., Gimeno-Sotelo, L. (2026). *Non-stationary dry-spell hazard probabilities for Spain.* International Journal of Climatology, e70393. https://doi.org/10.1002/joc.70393
3. El Kenawy, A. M., Domínguez-Castro, F., Beguería, S., Vela-Tambo, J., Halifa-Marín, A., Adell-Michavila, M., **Crespillo, A.**, Latorre, B., Franquesa, M., Vicente-Serrano, S. M. (2026). *Multiscalar and nonlinear controls of drought impacts in Spain revealed by media-reported data.* Natural Hazards and Earth System Sciences, 26(10), 4977–4996. https://doi.org/10.5194/nhess-26-4977-2026
4. Vicente-Serrano, S. M., Beguería, S., Reig, F., Royo, A., Arretxea, M., Gil, M., Latorre, B., El Kenawy, A., Franquesa, M., Halifa-Marín, A., Adell-Michavila, M., **Crespillo, A.**, et al. (2025). *Developing science-informed maps and climate service for extreme rainfall in Spain.* Natural Hazards, 121, 23281–23302. https://doi.org/10.1007/s11069-025-07731-0

5. Vicente-Serrano, S. M., Reig, F., Beguería, S., El Kenawy, A., Domínguez-Castro, F., Franquesa, M., Gimeno-Sotelo, L., Adell-Michavila, M., Halifa-Marín, A., Noguera, I., Andrés-Martín, M., Azorin-Molina, C., **Crespillo, A.**, Pérez-Pajuelo, D. (2025). *An optimal and flexible approach for drought quantification based on standardised indices.* International Journal of Climatology, 45(10), e8906. https://doi.org/10.1002/joc.8906
6. Halifa-Marín, A., et al., including **Crespillo, A.** (2025). *Spatial assessment of dry-spell hazards in Spain: toward an operational climate service.* Water Scarcity and Drought, 1(1), 4. https://www.sciltp.com/journals/wsd/articles/2509001429

IN PREPARATION

- **Crespillo López, A.**, et al. *Recent changes in hydrological drought propagation in river networks worldwide.*
- **Crespillo López, A.**, Michalek, A., et al. *Observed and simulated hydrological drought propagation with the distributed model Tiger-HLM over the United States.*

CONFERENCES

- **Crespillo López, A.**, Vicente-Serrano, S. M., Gimeno, L. (2026). *Event-based analysis of hydrological drought propagation across contrasting river basins.* EGU General Assembly, Vienna, May 2026. https://doi.org/10.5194/egusphere-egu26-1999
- **Crespillo López, A.** (2026). *Tracking hydrological drought propagation in river networks: an event-based framework.* International Conference on Hydroinformatics (HIC 2026), Zaragoza, June 2026.
- Vicente-Serrano, S. M., et al., including **Crespillo, A.** (2026). *Non-stationary probabilities of dry-spell hazard in Spain.* EGU General Assembly, Vienna, May 2026. https://doi.org/10.5194/egusphere-egu26-3914
- Trapero, L., **Crespillo, A.**, Udina, M. (2024). *Thermal inversion analysis in the Andorra Central Valley and its relationship with pollutants and meteorological variables.* European Meteorological Society (EMS) Annual Meeting. Work derived from the master's thesis. https://doi.org/10.5194/ems2024-722

OPEN SOURCE

- **Crespillo López, A.** (2026). *Tracking hydrological drought propagation in river networks: an event-based framework (code).* Zenodo, MIT licence. https://doi.org/10.5281/zenodo.22755834

TECHNICAL SKILLS

- Programming: Python and R. Analysis of hydrological and climate series, extreme-value statistics, machine learning.
- Machine and deep learning: scikit-learn, PyTorch, TensorFlow and Keras.
- Distributed hydrological modeling: Tiger-HLM, numerical experiments and evaluation with forcing ensembles.
- Spatial analysis: ArcGIS, drainage networks derived from digital elevation models.
- Computing: large-scale data processing, GPU use and high-performance computing.
- AI assistants applied to research: Claude (data analysis, code development and scientific writing).

Python · R · scikit-learn · PyTorch · TensorFlow · Keras · LaTeX · Git · Streamlit · Plotly · ArcGIS · GPU / HPC · Claude / AI

COURSES

- 17th Summer School on Machine Learning and Advanced Statistics, Technical University of Madrid.
- Bootcamp on data analysis and artificial intelligence, Upgrade Hub.
- IV Extremadura Congress of Climatology, University of Extremadura.
- Introduction to R and RStudio for data analysis in scientific research, Society of Friends of the National Museum of Natural Sciences.

LANGUAGES

- Spanish and Catalan: native languages.
- English: professional working proficiency. Duolingo English Test 120/160 (CEFR B2), October 2026 — subscores: writing 130, listening 120, speaking 115, reading 115. Cambridge First (B2); preparing Cambridge Advanced (C1).
- French: DELF B2.

ADDITIONAL INFORMATION

- Worked throughout my studies to fund them: customer service and warehouse work in the family business, sports instructor, waiter, kitchen assistant, referee and carpenter.
- Interests: sport, mountains, cooking and travel.